

Data base project

Online Voting System - SRS

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1. Entities

VOTER

ADMIN

ELECTION

CANDIDATE

VOTE

PARTY

CONSTITUENCY

RESULT

Attributes

VOTER

- voter_id (PK)
- name
- cnic (UNIQUE)
- password
- constituency_id (FK)
- email

ADMIN

- admin_id (PK)
- username
- password
- role
- email
- created_at

ELECTION

- election_id (PK)
- title
- election_date
- start_time
- end_time
- status

CANDIDATE

- candidate_id (PK)
- name
- party_id (FK)
- election_id (FK)
- age

- manifesto

VOTE

- vote_id (PK)
- voter_id (FK)
- candidate_id (FK)
- vote_time
- election_id (FK)
- status

PARTY

- party_id (PK)
- name
- symbol
- leader_name
- founded_year
- headquarters

RESULT

- result_id (PK)
- candidate_id (FK)
- election_id (FK)
- total_votes
- position
- result_status

Data Dictionary:

VOTER Data Dictionary

Column Name	Data Type	Size	Constraints	Description
voter_id	NUMBER	10	PK, NOT NULL	Unique voter ID
name	VARCHAR2	100	NOT NULL	Full name
cnic	VARCHAR2	15	UNIQUE, NOT NULL	National ID
password	VARCHAR2	100	NOT NULL	Login password
constituency_id	NUMBER	10	FK	Linked constituency
email	VARCHAR2	100	UNIQUE	Email address

ADMIN Data Dictionary

Column Name	Data Type	Size	Constraints	Description
admin_id	NUMBER	10	PK	Admin ID
username	VARCHAR2	100	NOT NULL	Admin username
password	VARCHAR2	100	NOT NULL	Password
role	VARCHAR2	50		Admin role
email	VARCHAR2	100	UNIQUE	Email
created_at	DATE	-		Creation date

ELECTION Data Dictionary

Column Name	Data Type	Size	Constraints	Description
election_id	NUMBER	10	PK	Election ID
title	VARCHAR2	100	NOT NULL	Election title
election_date	DATE	-	NOT NULL	Election date
start_time	DATE	-		Start time
end_time	DATE	-		End time

status	VARCHAR2	20	CHECK	Election status
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CANDIDATE Data Dictionary

Column Name	Data Type	Size	Constraints	Description
candidate_id	NUMBER	10	PK	Candidate ID
name	VARCHAR2	100	NOT NULL	Candidate name
party_id	NUMBER	10	FK	Linked party
election_id	NUMBER	10	FK	Linked election
age	NUMBER	3		Candidate age
manifesto	VARCHAR2	200		Manifesto

VOTE Data Dictionary

Column Name	Data Type	Size	Constraints	Description
vote_id	NUMBER	10	PK	Vote ID
voter_id	NUMBER	10	FK	Voter reference
candidate_id	NUMBER	10	FK	Candidate reference
vote_time	DATE	-		Time of vote
election_id	NUMBER	10	FK	Election reference
status	VARCHAR2	20		Vote status

PARTY Data Dictionary

Column Name	Data Type	Size	Constraints	Description
party_id	NUMBER	10	PK	Party ID
name	VARCHAR2	100	NOT NULL	Party name
symbol	VARCHAR2	50		Symbol
leader_name	VARCHAR2	100		Leader name

founded_year	NUMBER	4		Founded year
headquarters	VARCHAR2	100		Head office

CONSTITUENCY Data Dictionary

Column Name	Data Type	Size	Constraints	Description
constituency_id	NUMBER	10	PK	ID

RESULT Data Dictionary

Column Name	Data Type	Size	Constraints	Description
result_id	NUMBER	10	PK	Result ID
candidate_id	NUMBER	10	FK	Candidate reference
election_id	NUMBER	10	FK	Election reference
total_votes	NUMBER	10		Total votes
position	NUMBER	3		Rank
result_status	VARCHAR2	20		Status

2. Functional Requirements

Admin can create elections

Admin can add candidates

Admin can manage parties

Admin can view results

Voter can register

Voter can login securely

Voter can view candidates

Voter can cast vote

System prevents duplicate voting

System calculates results automatically

System displays winner

Admin can update election status

3. Non-Functional Requirements

System must ensure security through authentication and data protection

System should handle large number of users efficiently

System should be scalable for future expansion